Online Bayesian Learning Approach for Urban Roadbed Void Detection Using Vehicle-Induced Distributed Acoustic Sensing Signals

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***Abstract*—The imperative for urban infrastructure monitoring has catalyzed the development of Distributed Acoustic Sensing (DAS) technology. Addressing the specific near-field urban roadbed void detection and monitoring scenario, this paper proposes a method based on handling vehicle-induced signals as opportunistic sources. A physical model characterizing void-induced spatial coherence degradation and energy attenuation is established and online Bayesian learning algorithm is introduced to recursively estimate the statistics of features from streaming snapshots. Furthermore, spatial Constant False Alarm Rate (CFAR) detector and hypothesis testing theory are integrated to robustly localize the void by exploiting the spatial correlation in DAS array. Field experiments and quantitative analysis demonstrate precise void localization and robust detection performance of the proposed method, which offers a scalable solution for urban roadbed detection.**

***Index Terms*—Distributed acoustic sensing, online Bayesian learning, Spatial CFAR, roadbed void detection**

I. Introduction

With the rapid development of urbanization, the integrity of infrastructure has become paramount to both public safety and urban operational efficiency [1]. Precisely due to the pivotal role of urban roadbed, roadbed voids pose severe threats to people's lives, property and municipal functionality, making early monitoring and warning of urban roadbed void an urgent research imperative [2], [3]. Although the conventional nondestructive approach is capable of detecting urban roadbed void, it suffers from prohibitive costs and long-term monitoring inconvenience [4].

DAS is an emerging and cost-effective technology for monitoring tasks by transforming pre-existing optical fibers into massive ultra-dense sensor array with meter-level spatial resolution over kilometers [5], [6]. DAS has demonstrated strong potential for urban activity monitoring, ranging from transportation analysis to infrastructure assessment [2]. DAS-based sensing is extensively employed to monitor vehicle movements [7]–[9], estimate traffic speeds and weights [10], [11], and classify vehicle types [9], [12], [13]. Its utility extends to infrastructure structural health monitoring [14], [15] as well. DAS provides continuous spatial-temporal wavefield recording, making it an ideal candidate for multichannel signal processing tasks such as urban infrastructure monitoring [16].

Although DAS offers large-scale spatial coverage, inconsistencies in sensing sensitivity as well as ambient fluctuations [2] introduce significant noise and interference. These factors pose substantial challenges to the accuracy of signal processing and anomaly detection.

To mitigate the adverse effects of inconsistent sensor responses and fluctuating ambient noise in urban environments, some researchers have focused on leveraging diverse signal sources, which can be broadly categorized into active sources and opportunistic sources [17], depending on their excitation mechanism. Yao [3] utilized active-source signals generated by ground impacts to strengthen the signal intensity and achieved high spatial resolution void detection with reduced costs.

The practical implementation active-source excitation is often constrained by inherent operational infeasibility and long-term monitoring inconvenience. Hence, moving vehicles were proposed to serve as opportunistic sources and transform pre-existing fibers into sensor array via DAS technology [18], [19]. In the context, vehicle-induced signals refer to the dynamic strain and acoustic vibrations captured by the fiber-optic cable, which are activated by the vehicles traveling on or near the sensor array. Methods utilizing opportunistic sources facilitate sustained monitoring and significantly reduces computational overhead compared to conventional ambient noise interferometry, albeit at the cost of higher processing difficulty. Zhang [20] leveraged multiple effective features of vehicle-induced signals and trained LSTM-AE model to exploit temporal dependencies and localize voids. Integrated DAS/DTS system [21] synthesized vehicle-induced opportunistic vibration signal and temperature information, demonstrated the effectiveness for manhole localization and cable status anomaly detection.

While their methods demonstrate remarkable effectiveness, the inherent black-box nature of the model poses certain challenges regarding interpretability. Consequently, in an ini-

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tial attempt to enhance physical interpretability, we introduce a physics-guided array processing framework. By modeling roadbed void detection as a hypothesis test problem and leveraging opportunistic vehicle-induced signals to recursively estimate the statistics, the Spatial CFAR [22] based method yields encouraging initial results in roadbed void localization.

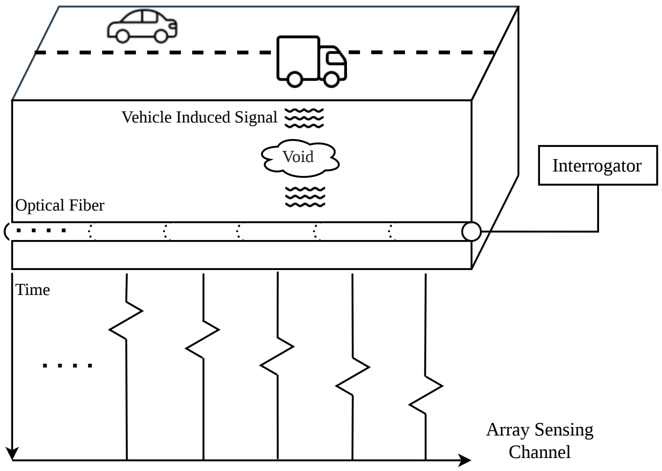


Fig. 1: Schematic of the DAS-based monitoring system.

Notation: $\mathbf{M}^\top$ and $\mathbf{M}^{-1}$ are the transpose and inverse of a matrix $\mathbf{M}$. $|\cdot|$ denotes the cardinality operator and $\|\cdot\|$ denotes the $\mathbf{L}_2$ norm operator. $\mathrm{std}(\cdot)$ and $\mathrm{var}(\cdot)$ denote the standard deviation and variance operators, respectively. $\oplus$ denotes the Minkowski sum operator.

II. Physical Model

This section introduces the DAS array deployment scenario, followed by the mathematical representation of vehicle-induced opportunistic sources, and the theoretical derivation of void-induced impact on acquired signals.

Fig. 1 illustrates the schematic of the DAS-based urban roadbed monitoring system. As a vehicle traverses the roadway, the vehicle-induced perturbations are captured by DAS sensor array. Simultaneously, phase information is demodulated into signals over multiple array sensing channels through an interrogator [23], [24], which is positioned at the fiber terminus.

Consider a DAS system deployed as a Uniform Linear Array (ULA) with N elements along the road. Let $\mathbf{X} \in \mathbb{R}^{N\times T}$ denote the received array signal, where N is the number of sensing channels, T is the observation duration, and $\mathbf{X}_{n,t}$ represents the strain measurement at sensing channel n and global time index t.

Due to the propagation kinematics, the vehicle-induced signal arrival time varies across sensing channels. Define a time-alignment vector $\boldsymbol{\tau} = (\tau_1, \ldots, \tau_N)^\top$, where τ_n is the time-of-arrival at sensing channel n. To capture the local dynamic strain, a temporal window of length L is extracted starting at τ_n. The aligned segment $\mathbf{x}_n \in \mathbb{R}^L$ is defined as

$$\mathbf{x}_n = \big[x_n(\tau_n),\, x_n(\tau_n+1),\, \ldots,\, x_n(\tau_n+L-1)\big]^\top, \quad (1)$$

and the trajectory-aligned signal matrix $\mathbf{T} \in \mathbb{R}^{L\times N}$ as

$$\mathbf{T} = \big[\mathbf{x}_1, \ldots, \mathbf{x}_N\big]. \quad (2)$$

$\{\mathbf{T}_m\}_{m=1}^{M}$ corresponds to set of trajectories, and prior trajectory set $\mathbf{\Phi}_0 = \{\mathbf{T}_m\}_{m=1}^{K}$ is used to estimate the initial parameters of the statistical model. The vehicle-induced DAS signals consist of quasi-static signals and propagating surface waves, the later is primarily comprised of Rayleigh waves [7], [25]. Formally, the channel signal is decomposed as

$$\mathbf{x}_n(t) = \mathbf{x}_n^{(\mathrm{q})}(t) + \mathbf{x}_n^{(\mathrm{r})}(t) + \mathbf{w}_n(t), \quad (3)$$

where $\mathbf{x}_n^{(\mathrm{q})}(t)$ and $\mathbf{x}_n^{(\mathrm{r})}(t)$ denote the quasi-static and Rayleigh signals, respectively, and $\mathbf{w}_n(t)$ denotes Additive White Gaussian Noise (AWGN).

Based on physical modeling, in the presence of subsurface void, the Soil Arching Effect redistributes the stress induced by vehicle loads from the region above the void to the surrounding stable structure [26]. The redistribution of stresses that ensues from soil arching leads to stress reduction in the yielding zone. Consequently, the low-frequency quasi-static signals perceived by the optical fiber will exhibit significant attenuation. The interaction between vehicle movement and the quasi-static strain induces edge effects around the void. These distortions compromise the signal integrity, resulting in a noticeable drop in the correlation coefficient at the transition zones.

III. Methodology

A. Statistical Modeling

To quantitatively estimate the mean levels and evaluate the degree of deviation, we formulated a statistical model to characterize the theoretical properties of the extracted features.

Let $\varepsilon_0 = (1/N_s)\sum_{i=1}^{N_s} x(i)^2$ denote the power of the composite signal. The background noise energy typically follows a Chi-square distribution [27], which asymptotically converges to a Gaussian distribution as $N_s \to \infty$ via the Central Limit Theorem. Logarithmic transformation, $\varepsilon = \ln(\varepsilon_0)$, is applied to suppress the heavy tail and enhance the symmetry of the distribution to adapt ε to be effectively approximated as a Gaussian statistical variable.

Under the high SNR conditions observed during vehicle passage, classical estimation theory [28] suggests that for a sufficiently large window length N_s, the sampling distribution of the correlation estimator for the spatial correlation coefficient ρ between DAS channels exhibits asymptotic normality. Finally, the joint feature vector $\mathbf{f} = [\varepsilon, \rho]^\top$ is modeled as a multivariate Gaussian variable $\mathbf{f} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{\Sigma})$ to facilitate computationally efficient, closed-form sequential updates within the proposed Bayesian framework.

B. Frequency and Spatial Features

Due to the physical effects described in Section II, quasi-static signal energy decays over void regions, leading to reduced SNR in central channels and decreased spatial correlation at the transition edge zones. Accordingly, we define the n-th channel feature vector $\mathbf{f}_n \in \mathbb{R}^D$ as

$$\mathbf{f}_n = \big[\mathrm{SNR}_n,\, Q_n,\, \mathrm{Corr}_n,\, \mathrm{Edge}_n\big]^\top, \quad (4)$$

Algorithm 1 Online Bayesian Detection with Spatial CFAR

Require: $\mathbf{X} \in \mathbb{R}^{T\times N}$; $\{\mathbf{T}_m\}_{m=1}^{M}$; $\{\mathbf{\Omega}_n\}_{n=1}^{N}$; $\mathbf{\Phi}_0$.
Ensure: Statistics $\mathbf{Z_m}$; Posterior $\mathbf{\Theta}_m$.
  Initialize: $\mathbf{\Theta}_0 = \{\boldsymbol{\mu}_0, \beta_0, \mathbf{W}_0, \nu_0\}$ via priors $\mathbf{\Phi}_0$.
  while stream is active **do**
    1. Feature: $\mathbf{f}_m = \mathcal{F}(\mathbf{T}_m)$
    2. Update $\mathbf{\Theta}_m$**:** $\beta_m = \beta_{m-1} + 1, \quad \nu_m = \nu_{m-1} + 1$
    $\boldsymbol{\mu}_m = (\beta_{m-1}\boldsymbol{\mu}_{m-1} + \mathbf{f}_m)/\beta_m$
    $\mathbf{W}_m^{-1} = \mathbf{W}_{m-1}^{-1} + \frac{\beta_{m-1}}{\beta_m}(\mathbf{f}_m - \boldsymbol{\mu}_{m-1})(\mathbf{f}_m - \boldsymbol{\mu}_{m-1})^\top$
    3. Spatial CFAR:
      Derive background $\Psi_{n,m}(\boldsymbol{\mu}_\Psi, \mathbf{\Sigma}_\Psi)$*:*
      $\boldsymbol{\mu}_\Psi = \frac{1}{|\mathbf{\Omega}_n|}\sum_{j\in\mathbf{\Omega}_n} \boldsymbol{\mu}_{j,m}$
      $\mathbf{\Sigma}_\Psi = \frac{1}{|\mathbf{\Omega}_n|}\sum_{j\in\mathbf{\Omega}_n} (\boldsymbol{\mu}_{j,m} - \boldsymbol{\mu}_\Psi)(\boldsymbol{\mu}_{j,m} - \boldsymbol{\mu}_\Psi)^\top$
      Compute anomaly score:
      $\mathbf{Z}_{n,m} = (\mathbf{f}_{n,m} - \boldsymbol{\mu}_\Psi)^\top \mathbf{\Sigma}_\Psi^{-1}(\mathbf{f}_{n,m} - \boldsymbol{\mu}_\Psi)$
    $m \leftarrow m + 1$
  end while
  return $\{\mathbf{Z}_m\}_{m=1}^{M}, \{\mathbf{\Theta}_m\}_{m=1}^{M}$

and features in $\mathbf{f}_n$ are defined as

$$\mathrm{SNR}_n = 10\log_{10}\left(\frac{\|\mathbf{x}_n\|^2}{\|\mathbf{w}_n\|^2}\right), \tag{5a}$$

$$Q_n = 20\log_{10}\left(\frac{\max_{p\in\mathcal{K}_q} P_n[p]}{\mathrm{std}(\mathbf{w}_n)}\right), \tag{5b}$$

$$\mathrm{Corr}_n = \frac{1}{|\mathbf{\Omega}_n^{in}|}\sum_{j\in\mathbf{\Omega}_n^{in}} \frac{\mathbf{x}_n^\top(\tau_n)\mathbf{x}_j(\tau_n)}{\|\mathbf{x}_n(\tau_n)\|\cdot\|\mathbf{x}_j(\tau_n)\|}, \tag{5c}$$

$$\mathrm{Edge}_n = \frac{1}{K-k}\sum_{k\le j<K} \frac{\mathbf{x}_{n-j}^\top(\tau_n)\mathbf{x}_{n+j}(\tau_n)}{\|\mathbf{x}_{n-j}(\tau_n)\|\cdot\|\mathbf{x}_{n+j}(\tau_n)\|}, \tag{5d}$$

where k and K denote the inner and outer neighborhood radius respectively, $\mathcal{K}_q \subseteq \mathbb{R}$ denotes the set of discrete frequency values corresponding to the quasi-static signal band, $P_n[p]$ represents the spectral amplitude at channel n and frequency value p. A general symmetric neighborhood set $\mathcal{N}_n(r)$ centered at channel n with a radius r was defined as

$$\mathcal{N}_n(r) = \{i \mid \max(1, n-r) \le i \le \min(N, n+r)\}, \tag{6}$$

the near neighborhood Corr_n can be formulated through the set minus operator $\setminus$ as follows:

$$\mathbf{\Omega}_n^{\mathrm{in}} = \mathcal{N}_n(k) \setminus \{n\}. \tag{7a}$$

C. Online Bayesian learning

To adapt to dynamic environments without retaining excessive historical snapshots, an online Bayesian approach was adopted to estimate the statistics of each sensing channel. The background statistics of the feature vector $\mathbf{f_n}$ are modeled using a multivariate Gaussian distribution with unknown expectation and covariance. The application of a Normal-Wishart conjugate prior ensures a computationally friendly update step without data retention [29]. Posterior from previous iteration recursively serves as the prior for the current step, written as

$$p(\mathbf{f}_m|\boldsymbol{\mu}_m, \mathbf{\Lambda}_m) \propto p(\boldsymbol{\mu}_m, \mathbf{\Lambda}_m|\mathbf{f}_m)p(\mathbf{f}_m|\boldsymbol{\mu}_{\mathrm{m-1}}, \mathbf{\Lambda}_{\mathrm{m-1}}). \tag{8}$$

Under the Bayesian recursive framework, the posterior distribution preserves the Gaussian-Wishart conjugate structure:

$$p(\mathbf{f}, \mathbf{\Lambda}|\boldsymbol{\mu}, \beta, \mathbf{W}, \nu) = \mathcal{N}(\mathbf{f}|\boldsymbol{\mu}, (\beta\mathbf{\Lambda})^{-1})\mathcal{W}(\mathbf{\Lambda}|\mathbf{W}, \nu), \tag{9}$$

where $\mathbf{\Lambda} \triangleq \mathbf{\Sigma}^{-1}$ is the precision matrix, while the hyper-parameters $\mathbf{\Theta} = \{\boldsymbol{\mu}, \beta, \mathbf{W}, \nu\}$ represent the mean estimate, belief confidence (equivalent observations), scale matrix, and degrees of freedom, respectively. The resulting closed-form update equations are detailed in Step 2 of **Algorithm 1**, based on derivations from [30].

D. Anomaly Detection as Hypothesis Test

To guarantee robustness, we employ Spatial CFAR detector utilizing the spatial filtering mechanism. Equation 6 defines $\mathbf{\Omega}_n$ as the neighborhood set for the n-th channel, whose parameters are configured according to the realistic spatial scale of the detection task, we model the background features as a multivariate Gaussian distribution:

$$p(\mathbf{f}_n \mid \boldsymbol{\mu}_n, \mathbf{\Sigma}_n, \mathbf{\Omega}_n) = \mathcal{N}(\mathbf{f}_n \mid \boldsymbol{\mu}_n, \mathbf{\Sigma}_n, \mathbf{\Omega}_n), \tag{10a}$$
$$\mathrm{d}^2(\mathbf{f}_n) = (\mathbf{f}_n - \boldsymbol{\mu}_n)^\top \mathbf{\Sigma}_n^{-1}(\mathbf{f}_n - \boldsymbol{\mu}_n), \tag{10b}$$

where $\mathrm{d}^2(\mathbf{f}_n)$ is the squared Mahalanobis distance [31] indicating deviation from the mean in D-dimensional space. Formulating void detection as a binary hypothesis test, the null hypothesis $\mathcal{H}_0$ corresponds to the absence of a void, while the alternative hypothesis $\mathcal{H}_1$ represents the presence of a void

$$\begin{cases} \mathcal{H}_0 : \mathbf{f}_n \sim \mathcal{N}(\boldsymbol{\mu}_n, \mathbf{\Sigma}_n), \\ \mathcal{H}_1 : \mathbf{f}_n \sim \mathcal{N}(\boldsymbol{\mu}_n + \mathbf{\Delta}\boldsymbol{\mu}_n, \mathbf{\Sigma}_n + \mathbf{\Delta\Sigma}_n). \end{cases} \tag{11}$$

Under $\mathcal{H}_0$, the test statistic after the m-th update, defined as $\mathbf{Z}_{m,n} = \mathrm{d}^2(\mathbf{f}_n)$, follows a Chi-squared distribution with $D = |\mathbf{f_n}|$ degrees of freedom. The detection threshold $\eta = \mathrm{F}_{\chi_D^2}^{-1}(1 - P_{\mathrm{fa}})$ is determined by a constant false alarm rate P_{fa}, where $\mathrm{F}_{\chi_D^2}^{-1}(\cdot)$ is the inverse cumulative distribution function, and the detection is performed according to the decision rule

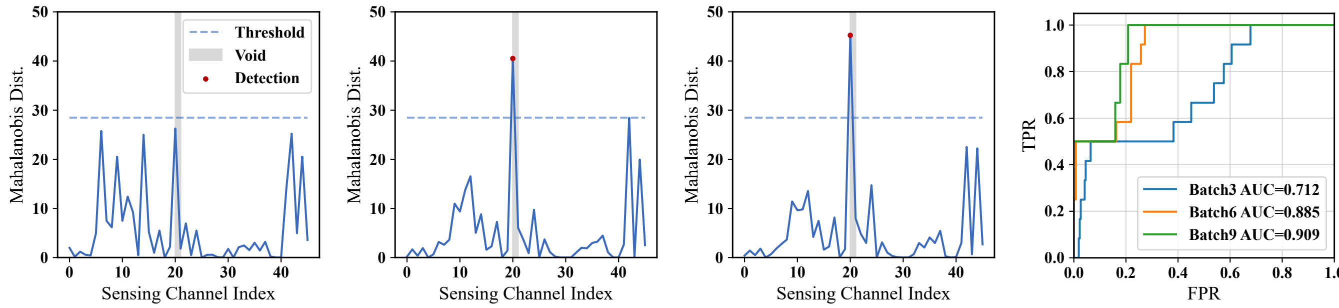
$$\mathbf{Z}_{n,m} = \mathrm{d}^2(\mathbf{f}_n) \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} \eta. \tag{12}$$

IV. Experiment Results

Field experiments were conducted on Daishan Road, Nanjing, China, between July and September 2024. Detailed experimental information is summarized in Table I.

TABLE I: Experimental Conditions

Condition	Detail
Void size	$1.5\,\mathrm{m} \times 1.5\,\mathrm{m} \times 1.0\,\mathrm{m}$
Overburden thickness	$0.1\,\mathrm{m}$
Fiber depth	$1.2\,\mathrm{m}$
Channel spacing	$0.8\,\mathrm{m}$
Sampling rate	$4\,\mathrm{kHz}$
Vehicle	Truck (20 t)
Average speed	$6.25\,\mathrm{m/s}$
Pavement type	Cement concrete
Environment	Sunny, $25 - 37\,°\mathrm{C}$



(a) Batch 3 (b) Batch 6 (c) Batch 9 (d) ROC

Fig. 2: Spatial CFAR detection performance across online Bayesian learning. (a)(b)(c) represent Mahalanobis Distance of Batch 3, 6, and 9. Therein $P_{\text{fa}} = 1 \times 10^{-4}$ (d) ROC evolution across samples accumulation.

Fig. 2(a)–(c) illustrate the convergence behavior after processing a series of vehicle trajectories. With the constant false alarm rate set to $P_{\text{fa}} = 1 \times 10^{-4}$, the proposed algorithm successfully localizes the anomalous region. This detection aligns precisely with the ground truth of the embedded void. Fig. 2(d) depicts the Receiver Operating Characteristic (ROC) curves and the corresponding Area Under the Curve (AUC) values for the respective recursive batches.

DAS geolocalization via GPS-tracked vehicles offers a precision of 1–5 meters [32] and tap test calibration is typically limited to a precision of 2–3 sensing channels [33]. Consequently, a spatial tolerance boundary was incorporated during the AUC analysis and quantitative assessment.

An ablation analysis was conducted on the features vector $\mathbf{f}_n$ to evaluate the contribution of individual feature, measured by AUC and FPR@99% (False Positive Rate at 99% Recall). The complete configuration achieves optimal AUC and FPR@99% performance. Conversely, omitting any single feature leads to a decline in both metrics. Notably, removing any of the SNR features results in the most severe performance degradation, underscoring their critical roles mediately.

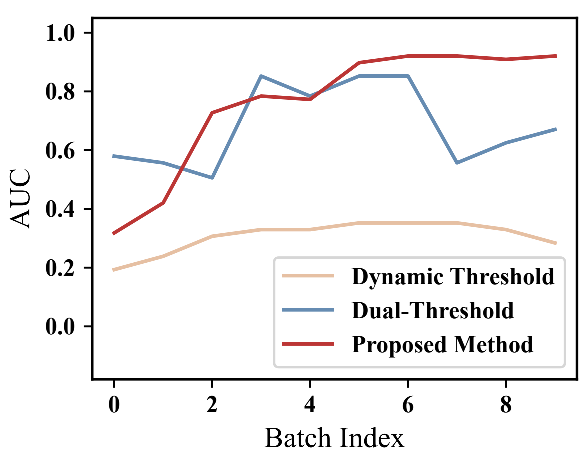


Fig. 3: Cumulative Sample AUC Analysis vs. Baselines.

Fig. 3 illustrates benchmark of the proposed framework against two baseline approaches, the conventional dynamic energy detector combined with wavelet denoising [34] and the dual-threshold approach [3]. In contrast, the performance curve of the proposed method exhibits a monotonic increasing tendency with sample accumulation and superior AUC scores, which demonstrate efficient convergence and robust stability in the post-convergence phase.

TABLE II: Quantitative Comparison

Method	**MarginIoU**
Dynamic Threshold [34]	0.19
Dual-threshold [3]	0.37
Ours(Nonrecursive)	**0.90**
Ours(Recursive)	**0.90**

For a quantitative assessment of the detection performance, we extended the Intersection over Union (IoU) metrics [35] MarginIoU, defined as

$$\text{IoU}_{margin} = \frac{|(\mathcal{P} \oplus \mathcal{B}_b) \cap (\mathcal{G} \oplus \mathcal{B}_b)|}{|(\mathcal{P} \oplus \mathcal{B}_b) \cup (\mathcal{G} \oplus \mathcal{B}_b)|} \tag{13}$$

where $\mathcal{P}$ and $\mathcal{G}$ denote the sets of predicted and ground truth channel indices, respectively, and $\mathcal{B}_b = \{-b, \dots, b\}$ is the structuring element defining the permissible tolerance zone. The quantitative analysis in Table II corroborates effectiveness in localization and consistent profile between recursive and nonrecursive Bayesian approach.

V. Conclusion

In this work, we propose a physics-guided array processing method for roadbed void detection using opportunistic vehicle-induced DAS signals. An online Bayesian learning algorithm is formulated to recursively estimate statistical distributions of features related to the physical degradation, enabling the precise identification of anomalies within non-stationary environments. Field experiments, ablation studies, and quantitative benchmarks validate the robust performance of the approach.

While the current field experiments were conducted under relatively controlled conditions involving specific vehicle types. Future research will comprehensively evaluate parameter sensitivity and focus on enhancing the method's adaptability across diverse vehicle types, weights, speeds, and more complex urban traffic environments.

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